

22PH202 – Electromagnetism and Modern Physics\

Unit II – Magnetism

2.1.Introduction about magnetism

Many of our modern technological devices rely on magnetism and magnetic materials. In olden days, magnets were used as magnetic compass for navigation, magnetic therapy for treatment and also used in magic shows. In modern days, many things we use in our daily life contain magnets, Motors, cycle dynamo, loudspeakers, magnetic tapes used in audio and video recording, mobile phones, head phones, CD, pen-drive, hard disc of laptop, refrigerator door, generator etc,

Magnetism is the force exerted by magnets when they attract or repel each other. Magnetism is caused by the motion of electric charges. Every substance is made up of tiny units called atoms. Each atom has electrons, particles that carry electric charges. Spinning like tops, the electrons circle the nucleus, or core, of an atom. Their movement generates an electric current and causes each electron to act like a microscopic magnet.



Figure 1. Magnetic attraction of bar magnet with iron particles

2.2 Origin of magnetic moment

An electron is a charged particle and there is a magnetic moment associated with its angular momentum. Because the electron in an atom may have two types of angular momentum, spin and orbital angular momentum, there are two sources of magnetic moment. These two magnetic moments can interact and give rise to shifts in the energies of the states of the atom which affect the appearance of the spectrum of the atom. The resulting shifts and splitting of lines is called the fine structure of the spectrum.

The macroscopic magnetic properties of materials are a consequence of magnetic moments associated with individual electrons. Each electron in an atom has magnetic moments that originate from two sources

2.2.1 An orbiting electron

Electron's orbital motion around the nucleus. Electron is considered as a moving charge (a small current loop), generating a very small magnetic field.

Magnetic moment is along the axis of rotation.

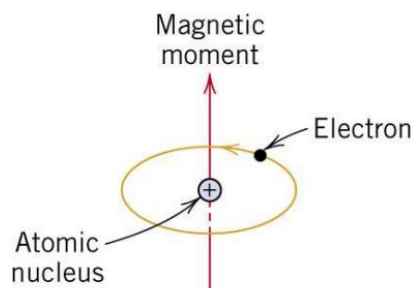


Figure 2. Orbital magnetic moment

2.2.2 A spinning electron

Each electron is thought to be spinning around an axis. Magnetic moment originates from this electron spin, which is directed along the spin axis. (Spin magnetic moments may be only in an “up” direction or in an anti-parallel “down” direction).

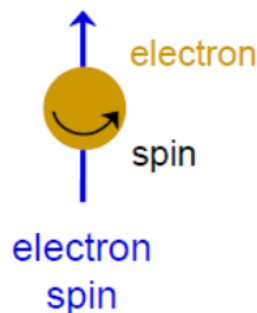


Figure 3. Spin magnetic moment

- Net magnetic moment is the sum of the magnetic moments of all the electrons including orbital and spin contributions.
- For an atom having completely filled electron shells or subshells, there is total cancellation of both orbital and spin moments.



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- Thus materials composed of atoms having completely filled electron shells are not capable of being permanently magnetized (Inert gases He, Ne, Ar and some ionic materials)
- For the electron spin, the ratio of magnetic moment to mechanical angular momentum is twice as large as for the orbital angular momentum of the electron.